INCH-POUND

A-A-59418
19 December 2003

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COMMERCIAL ITEM DESCRIPTION
METER, MODULATION
(150 kHz to 1.3 GHz)

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies

1. SCOPE. This Commercial Item Description (CID) describes a FM/AM Modulation meter. This instrument gives a complete accurate characterization of modulated signals in the frequency range of 150 kHz to 1.3 GHz. This CID is meant as a minimum requirement for the modulation meter, in which only those manufacturers that meet or surpass the following requirements are supplied per this CID.

2. SALIENT CHARACTERISTICS. The equipment shall be capable of operation within the accuracies, limits, and specifications herein. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be Class 3 in accordance with MIL-PRF-28800 except as specified herein.

2.3 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

2.4 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 2000 hours of operation with a statistical certainly of 95%.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611

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2.5 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater, based on an operating time of 2000 hours per year.

2.6 Electrical Power Sources and Connections. The equipment shall operate from nominal commercial, military, and shipboard power source of 120Vrms or 240Vrms single-phase AC (± 10 %) at line frequencies of 50 Hz and 60 Hz (± 5 %) as specified in MIL-PRF-28800.

2.7 Performance Characteristics.

2.7.1 General Description. The FM/AM Modulation meter shall be able to analyze a minimum frequency of 150 kHz up to and including the frequency of 1.3 GHz. The FM/AM Modulation Meter shall come fully calibrated so that it meets or exceeds all performance and testing requirements in a laboratory environment.

2.7.2 Radio Frequency (RF) Input.

2.7.2.1 Carrier Frequency Range. The FM/AM Modulation Meter shall analyze a carrier frequency range from 150 kHz to 1.3 GHz. For phase modulation the carrier frequency shall be from 10 MHz to 1.3 GHz.

2.7.2.2 The FM/AM Modulation Meter shall have automatic or manual front panel tuning.

2.7.2.3 Maximum Input. The FM/AM Modulation Meter shall be able to withstand an application of 35 Vrms (+25 dBm with a SWR less than 4 at frequencies from 150 kHz to 1.3 GHz) or 40 V DC without any harmful effects.

2.7.2.4 Input Impedance. The FM/AM Modulation Meter shall have the input impedance of 50 ohms nominal with an N type connector used.

2.8 Frequency Modulation (FM).

2.8.1 Amplitude Module (AM) Rejection shall typically be less than 20 Hz peak deviation measured in a 50 Hz to 3 kHz bandwidth for 50 % AM at 400 Hz or 1 kHz rate.

2.8.2 Residual FM shall be less than 8 Hz at 1.3 GHz and decreases linearly with frequency to less than 1 Hz at 100 MHz and below.

2.8.3 FM Deviation Range. Carrier frequencies 150 kHz to 5.5 MHz must have a maximum deviation of 40 kHz at rate of 30 Hz to 10 kHz. Carrier frequencies of 10 MHz to 1.3 GHz must have a maximum deviation of 400 kHz at rates of 30 Hz to 200 kHz.

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For clarification purposes, the specifications are.

Maximum Deviation	Carrier Frequency	Rates
40 kHz peak	150 kHz to 5.5 MHz	30 Hz to 10 kHz
400 kHz peak	10 MHz to 1.3 GHz	30 Hz to 200 kHz

2.8.3.1 Accuracy of Frequency Modulated Reading on the following ranges must be:

1) 250 kHz to 10 MHz ± 2% of reading ± 1 digit at 20 Hz to 10 kHz rate.
2) 10 MHz to 1.3 GHz ± 1% of reading ± 1 digit at 50 Hz to 100 kHz.

2.8.3.2 FM Demodulated Distortion Output shall be less than 0.1% Total Harmonic Distortion (THD at less than 0.1% volt peak).

2.9 Amplitude Modulation (AM).

2.9.1. AM Capabilities. The instrument shall be capable of measuring AM signals with significant levels of FM present by using selected AM settings and FM rejection circuitry.

2.9.2 Amplitude Modulation Depth range shall be from 0 to 90%.

2.9.2.1 AM Demodulated Output Distortion shall be less than 0.3% Total Harmonic Distortion (THD) at less than or equal to 50% depths.

2.9.2.2 Accuracy of Amplitude Modulation. The instrument shall provide the following accuracies at the specified carrier frequency, rate and depth.

Accuracy	Carrier Frequency	Rates	Depth
±2% + 1 digit	150 kHz to 10 MHz	50 Hz to 10 kHz	> 5% to < 95%
±3% + 1 digit	150 kHz to 10 MHz	30 Hz to 10 kHz	> 5% to < 95%
±2.5% + 1 digit	10 MHz to 1.3 GHz	30 Hz to 50 kHz	> 5% to < 95%
±3% + 1 digit	10 MHz to 1.3 GHz	30 Hz to 100 kHz	> 5% to < 95%

2.9.3 Residual AM shall be less than 0.02% for input levels of 30 mV or less with a 300 Hz to 3 kHz bandwidth filter or equivalent filter selected.

2.9.4 Audio Filters. The high low/pass filters with de-emphasis time constants shall be front panel controllable.

2.10 Phase Modulation. The instrument shall have phase modulation accuracy of ± 3% of reading ±1 digit for modulation rates from 200 Hz to 20 kHz, with deviations up to 400 radians.

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2.10.1 AM/FM Calibrator. The modulation analyzer shall have AM and FM modulation capability to the accuracies specified in subparagraph 2.10.1a through f, as follows:

a) AM Calibration Depth and Accuracy: 33.33% depth nominal, internally calibrated to an accuracy of ± 0.1%.
b) FM Calibration Deviation and Accuracy: 34 kHz peak deviation nominal, internally calibrated to an accuracy of ± 0.1%.
c) Output Connector: Front panel BNC.
d) Output Carrier Frequency: 10 MHz nominal.
e) Output Level: - 25 dBm minimum.
f) Modulation Rate: 10 kHz nominal.

2.11 Output. The FM/AM modulation meter shall come with an IEEE-488 interface bus. The instrument shall also have demodulated FM and AM outputs. All function shall be automatic and easily programmable.

2.11.1 Frequency. The FM/AM Modulation shall supply an IF output from at least 150 kHz to 2.5 MHz. The output voltage shall be no less than 0.010 Vrms into a 50 ohms load.

2.12 The instrument shall come with an IEEE-488 connector.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). These are consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials

4. PRODUCT CONFORMANCE PROVISIONS

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of ASTM-SI10, and all other requirements of this CID are met.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Sources of Documents.

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6.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.1.2 The Code of Federal Regulation (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.1.3 IEEE standards can be obtained from IEEE Customer Service Center, 445 Hoes Lanes, P.O. Box 1331, Piscataway, NJ 08855-1311, U.S.A. Phone: 800-678-4333 (Toll-Free, USA & Canada), 732-981-0060 (Worldwide), Fax: 732-981-9667. Website: www.ieee.org.

MILITARY INTEREST

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